

THE LARMOR LADS

Marine Towfish Equipped with Proton Precession Magnetometer to Locate Wreckage

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PROJECT OVERVIEW

Search and recovery operations in shallow waterways are costly and logistically complex. Our team designed a marine towfish – a towed underwater sensor platform to survey shallow waters for submerged metallic wreckage like downed aircraft and sunken vessels. The device would use a custom-built proton precession magnetometer (PPM) to detect disturbances in Earth's magnetic field caused by these sunken metals, with anomalies logged and GPS-tagged in real time.

Key Design Goals:

- Low-cost relative to commercial alternatives
- Shallow water capability
- Highly sensitive sensing and amplification system

Project Subsystems:

- Power Supply
- Polarization Circuit
- Signal Processing & Amplification
- Data Logging & GPS Integration
- Mechanical Enclosure Design

Motivation: Commercially available towfish systems can cost anywhere from \$12,000 to over \$80,000, which is an unrealistic expense for nonprofits working to locate MIA soldiers.

TECHNICAL OVERVIEW

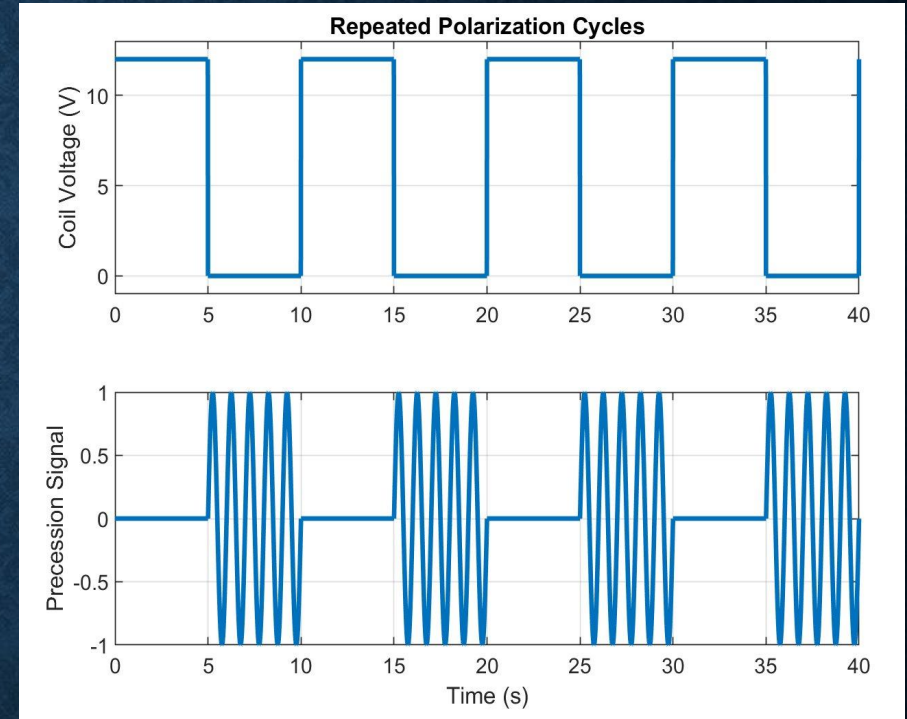
We ended up breaking our PPM into a couple major subsystems that will be detailed in the remainder of the presentation. Here is the basic overview about how the whole unit works to accomplish the goal:

Polarization – A strong DC magnetic field forces hydrogen protons into alignment within a distilled water sample

Precession – When the field is switched off, protons precess around Earth's ambient field (B_e) at the Larmor frequency: $f_L = \left(\frac{\gamma_p}{2\pi}\right) \cdot B_e$

Sensing – The precessing protons induce a small oscillating voltage in a pickup coil at exactly f_L

Singal Processing - By precisely measuring f_L , the instrument solves for $B_e = \frac{2\pi f_L}{\gamma_p}$, yielding an absolute scalar measurement with no external calibration needed.



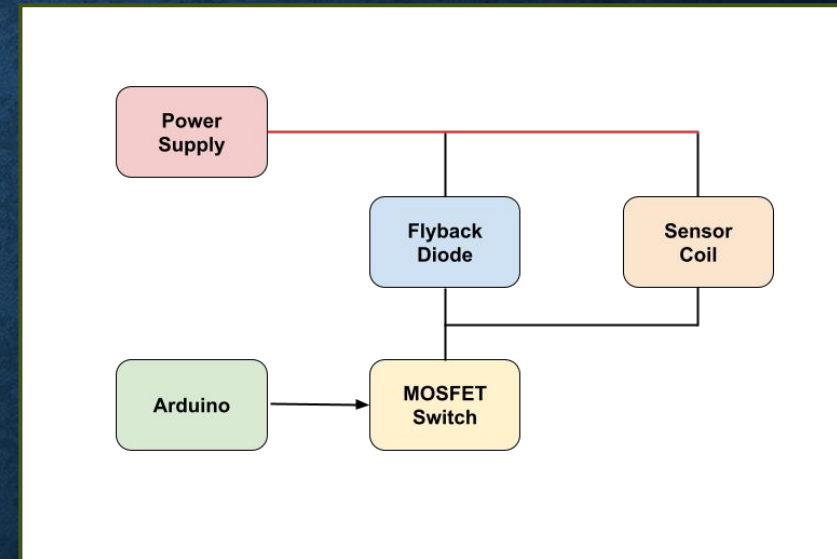
POLARIZATION DESIGN CHOICE – MOSFET SWITCHING

How it Works

- Arduino sends control signal → gate driver
- MOSFET acts as a low-side switch controlling current through the sensor coil
- ON: coil energized → magnetic field generated
- OFF: field collapses → precession begins
- Flyback diode protects against inductive voltage spikes

Why MOSFET?

- Handles high current efficiently
- Fast switching → precise timing control
- Low power loss compared to relays
- No mechanical wear → reliable operation



Key Specs: 12.8 V, 10 A, 5 V logic,
5s ON/5s OFF switching cycle

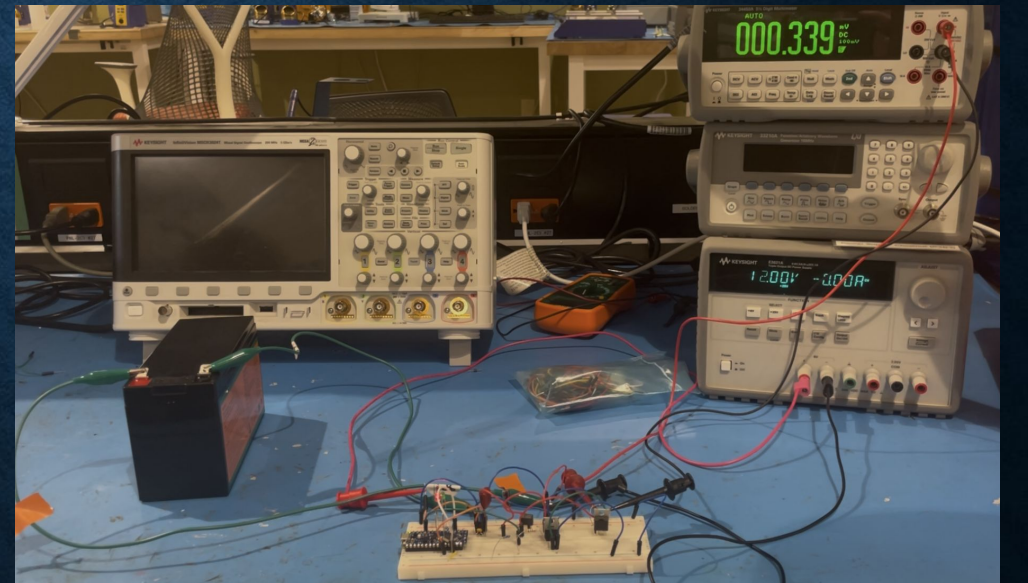
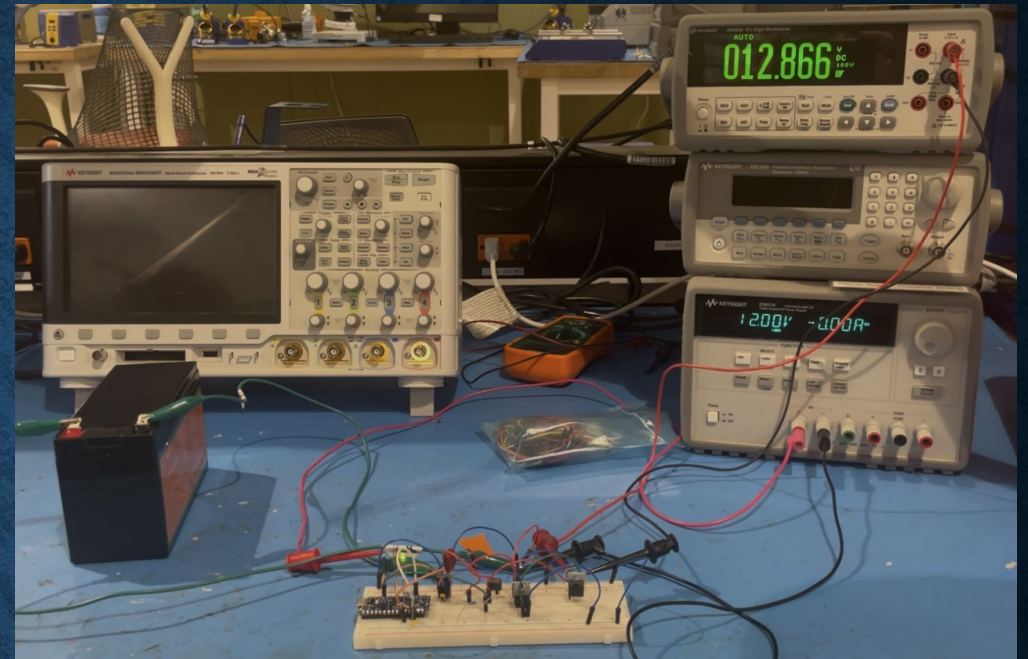
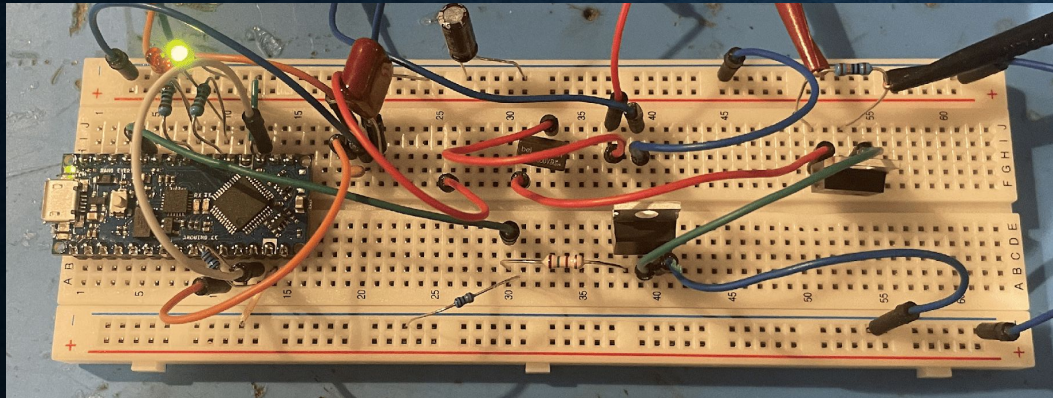
DEMO/LIMITATIONS

Demonstration

- Verified MOSFET switching controlled by Arduino
- Confirmed timing cycle (ON/OFF operation)
- Stable operation using 12.8 V battery
- Demonstrated safe switching of load

Limitations

- Breadboard current limit (1 A) prevents full-power testing
- Used resistor instead of coil to limit current
- Unable to test full polarization current
- High-current operation deferred to PCB implementation



HOW TO BUILD IT

(The Polarization Board)

IMPORTANT INFORMATION

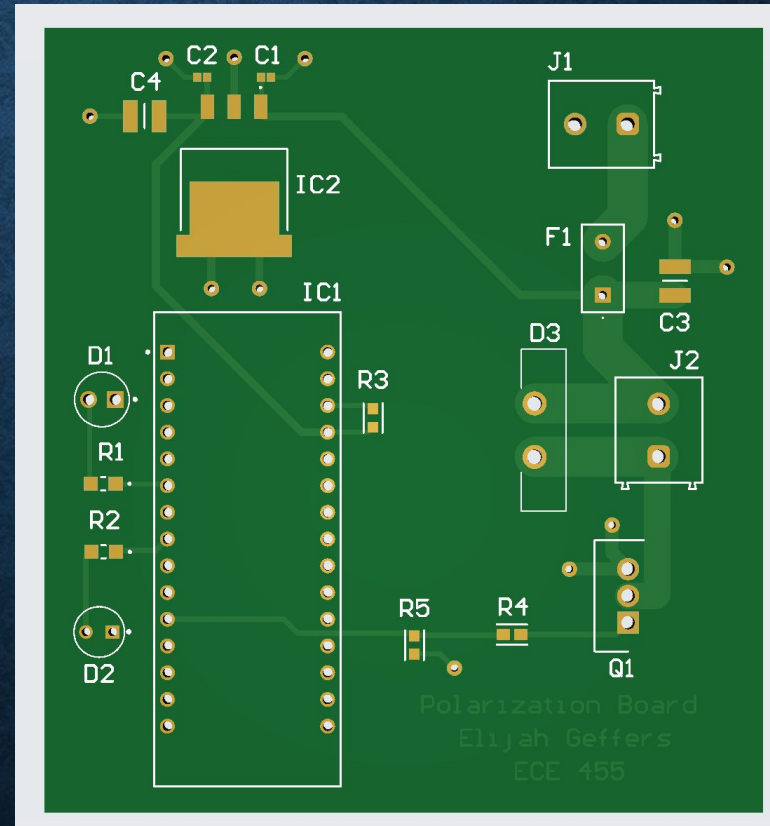
- Powered by a 12.8 V DC supply
- High-current path designed for up to 10 A
 - Coil current determined by: $I = 12.8 \text{ V} / R_{\text{coil}}$
 - Required coil resistance $\approx 1.3 \Omega$
- 5 V logic rail generated using a voltage regulator from 12.8 V input
- System divided into two functional sections:
 - Low-current logic (control & signaling)
 - High-current polarization circuit (coil drive)

GETTING STARTED

- Download all project files from GitHub
- Extract:
 - [PCB files](#) (.SchDoc, .PcbDoc)
 - [Gerbers](#) (for manufacturing)
 - [Arduino code](#) (.ino)
 - [BOM](#) (.csv)

PCB FABRICATION

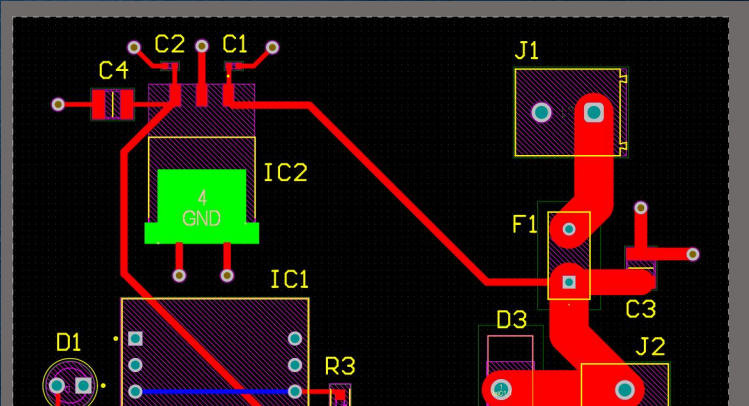
- Upload Gerber ZIP from GitHub to PCB manufacturer (JLCPCB recommended)
- Recommended settings:
 - 2-layer board
 - 1 oz copper
- FR-4 as base material

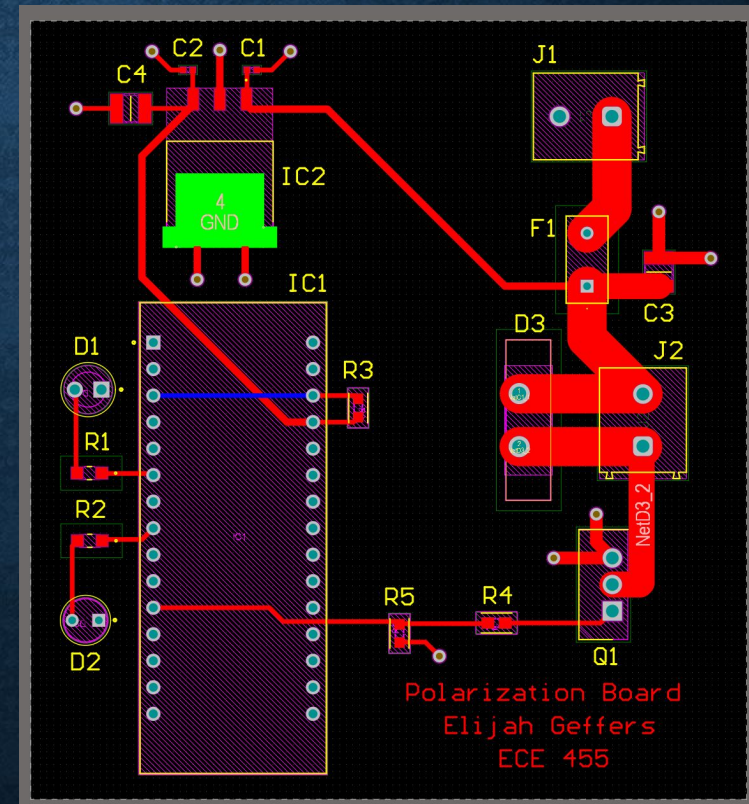


PCB ASSEMBLY

- Solder small components first (resistors, capacitors, fuse, LEDs)
- Install voltage regulator
- Solder MOSFET and flyback diode (make sure orientation is correct)
- Solder terminals for:
 - Power input
 - Coil output
- Solder Arduino last

WIRING AND CONNECTIONS

- Connect 12.8 V supply to input terminals (+ goes to the terminal connected to the trace)
 - Connect coil leads to output terminals
 - Connect Arduino:
 - Control pin (D8) → MOSFET gate
 - Make sure all grounds are shared
- 
- The diagram shows a PCB layout for a relay driver circuit. It includes an Arduino (IC1) connected to a MOSFET (IC2) which drives a relay (J1). The MOSFET's gate is connected to the Arduino's D8 pin. The MOSFET's drain is connected to the relay's coil (C1, C2, C3, C4). The MOSFET's source is connected to ground (GND). The relay's contacts (F1, F2) are connected to the output terminals (D1, D2). The circuit is powered by a 12.8V supply (J2) connected to the input terminals (D3, D4). A 10k resistor (R3) is connected between the MOSFET's gate and ground. The PCB is populated with various components including capacitors (C1, C2, C3, C4), a MOSFET (IC2), a relay (J1), and an Arduino (IC1).



PROGRAMMING THE ARDUINO

- Open Arduino IDE
- Load .ino file from GitHub
- Select correct board and port
- Upload code
- Default timing:
 - ~5s ON (polarization)
 - ~5s OFF (precession)

```
const int PIN_COIL = 8;    // MOSFET gate
const int PIN_GREEN = 3;   // Green LED
const int PIN_RED = 5;     // Red LED

const unsigned long ON_TIME = 5000;    // 5 seconds
const unsigned long OFF_TIME = 5000;   // 5 seconds

void setup() {
  pinMode(PIN_COIL, OUTPUT);
  pinMode(PIN_GREEN, OUTPUT);
  pinMode(PIN_RED, OUTPUT);

  // Start in OFF state
  digitalWrite(PIN_COIL, LOW);
  digitalWrite(PIN_GREEN, LOW);
  digitalWrite(PIN_RED, HIGH);
}

void loop() {
  // ON state
  digitalWrite(PIN_COIL, HIGH);    // energize coil
  digitalWrite(PIN_GREEN, HIGH);   // green LED ON
  digitalWrite(PIN_RED, LOW);      // red LED OFF
  delay(ON_TIME);

  // OFF state
  digitalWrite(PIN_COIL, LOW);     // turn off coil
  digitalWrite(PIN_GREEN, LOW);    // green LED OFF
  digitalWrite(PIN_RED, HIGH);     // red LED ON
  delay(OFF_TIME);
}
```


TESTING

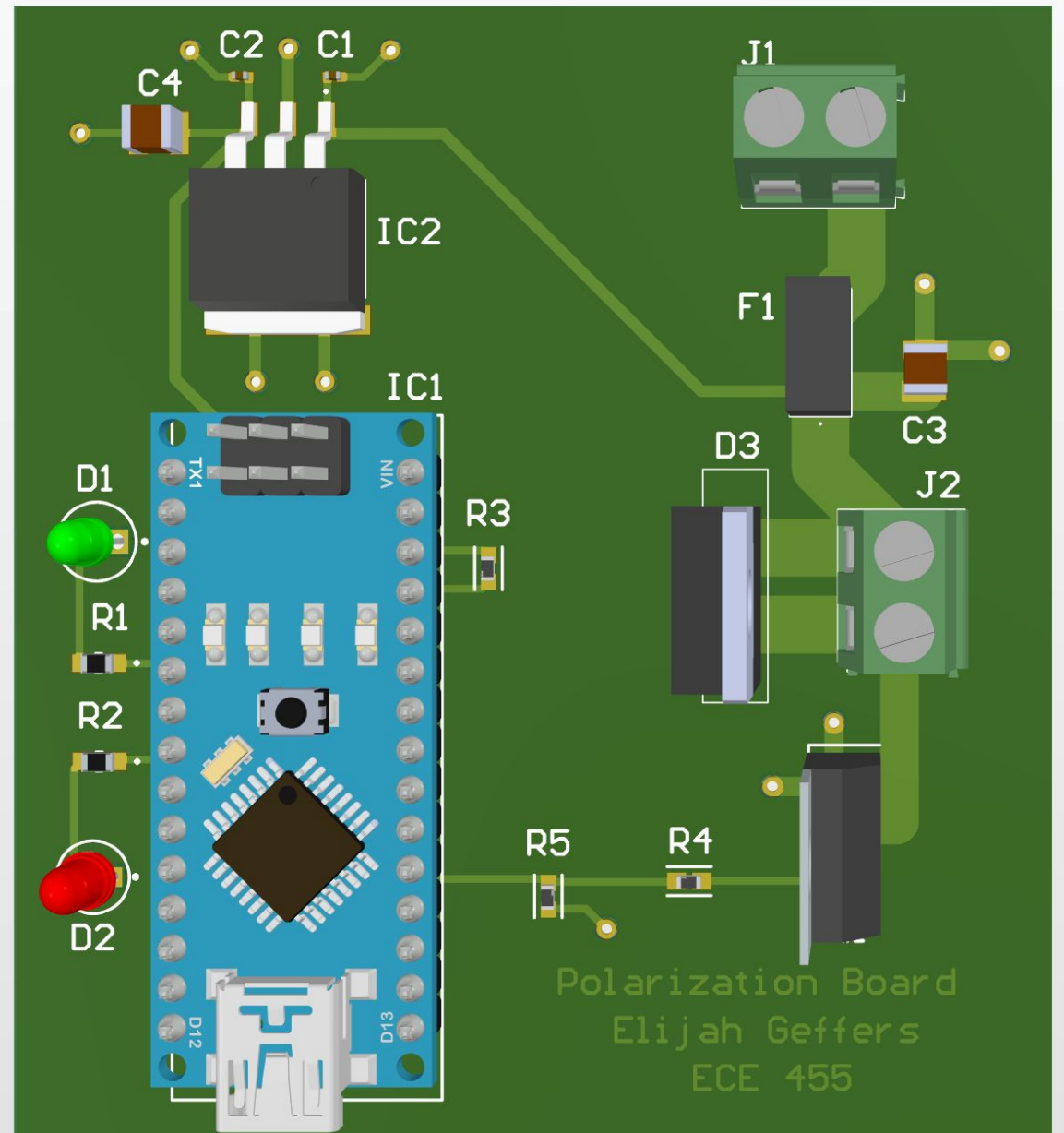
INITIAL

- Start with low current test ($\sim 1\text{ A}$)
- Use high-resistance load (not full coil)
- Make sure:
 - LEDs toggle correctly
 - MOSFET switching works
 - 5 V rail is stable

FINAL

- Apply full 12.8 V supply with coil
- System should cycle:
 - ON $\rightarrow$ coil energized (12.8 V, 10 A)
 - OFF $\rightarrow$ measurement phase (0 V, 0 A)
- Ensure:
 - No overheating
 - Stable switching

FINAL PRODUCT





COIL DESIGN OVERVIEW

- The magnetometer operates in two stages: polarization and sensing, using one or more coils for each step.
- There are many possible configurations for the coil, with many basic designs using two solenoids, some using dedicated polarization and sensing coils, and some using one large toroidal coil.
- A toroidal coil geometry was initially selected for the final design due to its superior angular stability compared to other configurations.
- We had to pivot due to design constraints to a 3 coil geometry for our final design (pictured), with a polarization coil surrounding two sensor coils.

CHOOSING THE COIL LAYOUT

- Initially, we believed a toroidal coil would be the best choice for the magnetometer, due to the large scope of research pointing to its advantages over standard solenoid designs.
- Prior research papers especially highlight the fact that toroidal coils are less sensitive to external noise, despite being difficult to build and holding less fluid.
- Another key difference is how it responds to orientation, even when a toroidal coil is aligned in the worst possible configuration, it can still produce half its maximum signal. A solenoid, on the other hand, produces no signal when it's exactly perpendicular to the field.
- Since it is less sensitive to orientation, the toroidal design initially seemed like a good fit for a towfish system, where the sensor trails behind a boat and constantly shifts angles.
- While we eventually used a 3 coil setup for the coils, the potential advantages of a toroidal coil for marine applications should not be ignored, and by adequately accommodating the difficult winding and wire constraints, the toroid could work in larger scale applications.

COIL DESIGN CHALLENGES

- The main limitations come from the unique geometry and wire necessary to increase the signal from the coil(s). The coil(s) need to have a large number of turns and have a long and narrow design but would also be small enough to fit inside a towfish.
- Additionally, the polarization stage relies on a strong current through the coil, making it ideal to minimize resistance, while the sensing stage relies on the number of turns, increasing resistance. This makes a Toroid especially difficult since maximizing number of turns while simultaneously minimizing resistance gets very expensive and impractical very quick.
- While designing the amplifier and polarization circuits, the calculations made it clear that our original Toroidal configuration was not going to be feasible; The toroid would have to be extremely large, have an excessive number of turns (~5000), and use very thick wire on account of being both the polarization and sensing coil. On top of this, it would have been extremely difficult to wind, requiring the full coil of wire to pass through the center of the toroid for every turn of wire.
- These problems could be slightly mitigated by modifying the toroid's structure and cross section, but still could not provide a strong enough signal to the amplifier circuit without increasing the power past safe levels for the polarization circuit.

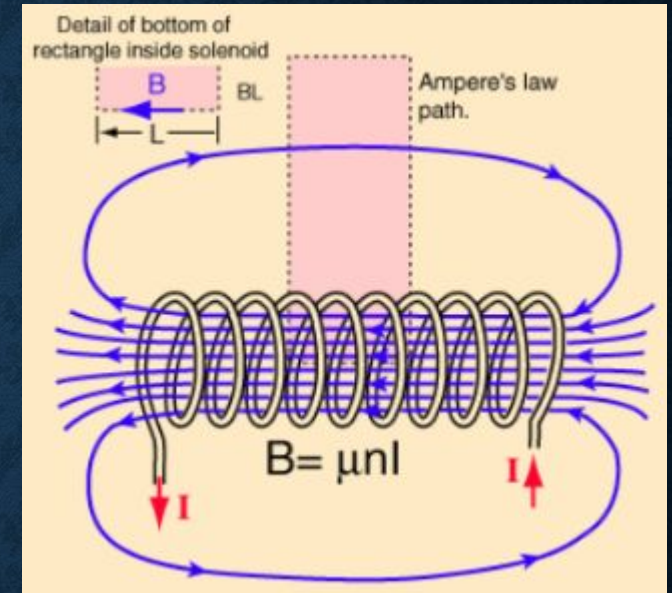


Figure 1.1 from "Proton Precession Magnetometers, Revision 3", James A. R. Koehler, March 2012, Highlighting the direction in which a coil should face (ensure the polarizing field is not in the same axis as the external field).

Toroid Volume (mL)	1542.125688
Length of Wire Coil Approx (m)	942.4777961

Calculations performed based on research by James A. R. Koehler, "Proton Precession Magnetometers, Revision 3", March 2012. Relevant Excel formula sheet can be found in Github and at [this link](#). - Tyler Fisher

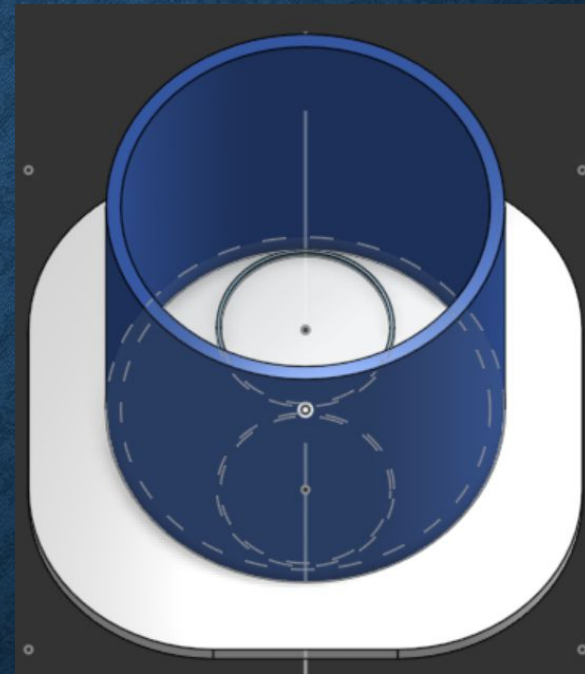
COIL DESIGN CHOICES

- Due to the limitations of the toroidal configuration previously mentioned, we decided to pivot to a 3 coil setup: Polarization, Sensor, and Cancellation, as there was ample research describing the process of construction in this configuration

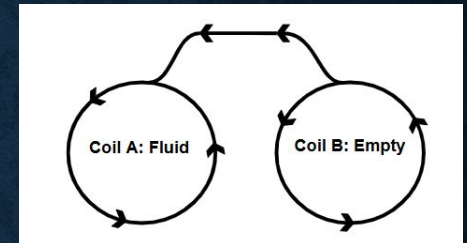
- The Polarization coil surrounds the other two coils, uses thick wire to maintain a low resistance while having enough turns to create a strong magnetic field within the Sensor coil's fluid.

- The Sensor and Cancellation coils have significantly more turns of thinner wire. Crucially, the two coils have nearly identical construction (Resistance and Inductance) to ensure optimal cancellation when wired in opposing configurations.

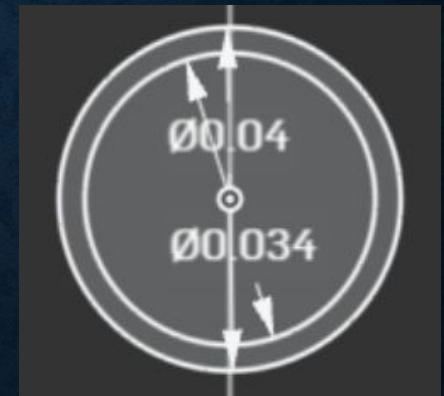
- Under this configuration, two outputs can be sent from the Sensor and Cancellation coils to a differential amplifier to ideally read the true signal ($\text{Signal} + \text{Noise from Coil A} - \text{Noise from Coil B}$).



Polarization coil design with baseplate and ridges for sensor coils
-Tyler Fisher



Coil Configuration - Tyler Fisher



Sensor coil measurements, relevant STL files can be found through Github - Tyler Fisher

PLANNING AND DESIGNING THE COILS

- Based on our research, we discovered that a 1 nT sensitivity required an ideal signal to noise ratio of ~ 100 .
- Our target for the amplification circuit was a signal strength on the order of microvolts, requiring very sensitive coils
- We were also limited by the polarization circuit, which had components capped at 10 Amps, and the availability of batteries limiting us to just 12 Volts
- Following the calculations based on our research in [this excel document](#), we settled on a 1.2 ohm polarization coil with 15 AWG wire, coming out to ~ 300 turns, with sensor coils approaching ~ 750 turns each.
- The resistance of the sensor coils is not as necessary to limit, so 22 AWG was used. More importantly, the two coils should be identical, which can be ensured by measuring and equating their resistances and inductances by trimming excess wire.

Polarization Coil

- Wide radius solenoid surrounding both sensor coils
- Driven by a pulsed polarization circuit
- Requires high current, resistance should be minimized
- Generates a strong magnetic field to align proton spins within the fluid

Sensor Coils

- Long, narrow solenoids with many turns to maximize induced signal
- One fluid filled, one empty for differential noise cancellation
- Detects decaying precession signal after polarization is removed
- Signal sent to amplifier

PRINT AND ASSEMBLY

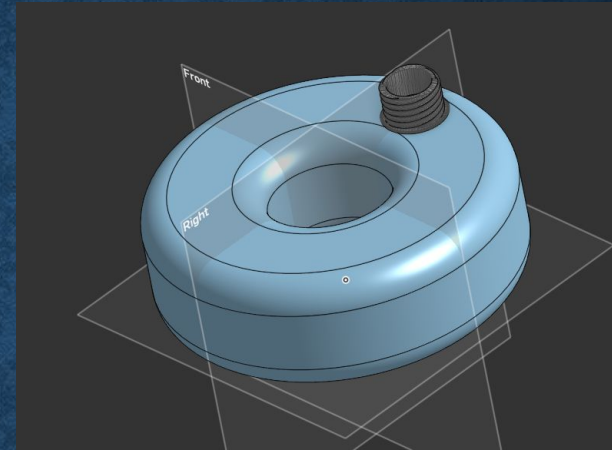
Once we decided the specifications of the coils including the geometry and number of turns, we began designing the coils

To create 3D models from scratch, we used Onshape, which has easy online access and ample tutorials. This allowed us to easily create many iterations of toroidal and solenoidal coils to procedurally test

To print the coils, we created slice files in Bambu Lab connected to 3D printers. It was crucial to enable supports for toroidal coils (see image), otherwise the coils would not properly print

Once printed, we removed any supports and wound the coils using the correct gauge of wire, counting the turns to ensure we met specification guidelines

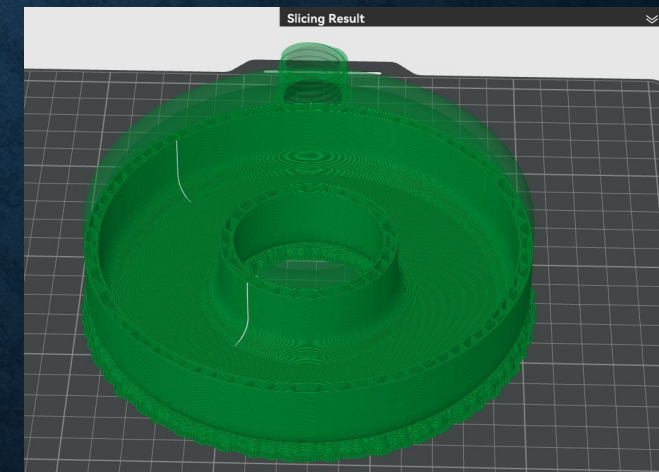
Finally, once winding was complete, we assembled the coils by placing them in the correct configuration and fastening any detached parts using adhesive



Example of a 3D toroid model produced with Onshape – this prototype had a cap and lid so the coil could be filled and sealed



Support settings for printing toroidal coils – notice the wide base in the slice file



Example slice file of toroidal coil from Bambu Lab –Tyler Fisher

AMPLIFICATION CIRCUIT

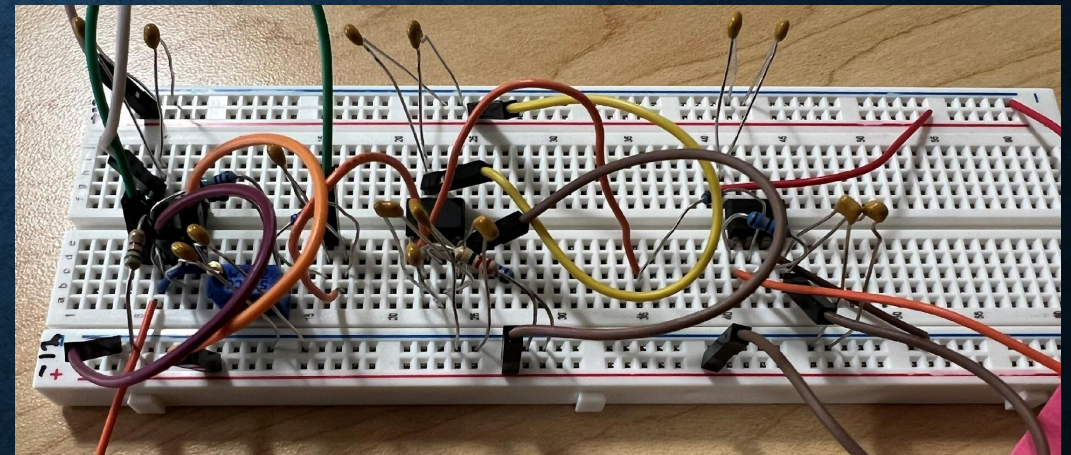
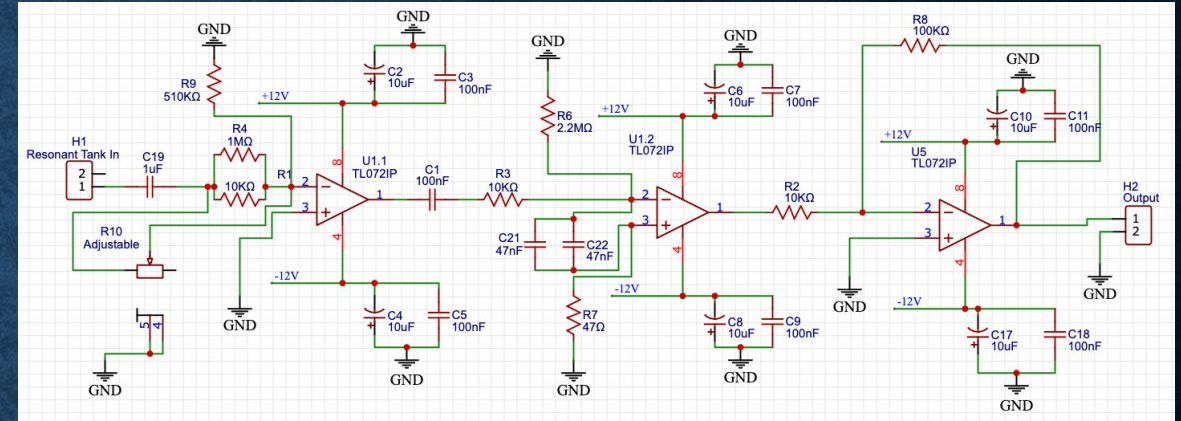
Goal: Amplify the weak ~ 2.1 KHz precession signal induced in the pickup coil for downstream frequency measurement.

Custom Design:

- 3-stage TL072IP op-amp chain with adjustable input gain
- ± 12 V decoupled power rails with bandpass filtering between stages
- Designed specifically around the expected Larmor frequency range

Why it Failed:

- The breadboard noise floor was high enough to completely mask the precession signal
- Power supply ripple fell directly within our target frequency band, making filtering ineffective
- Inductive coupling from the polarization circuit was corrupting the input before amplification even began

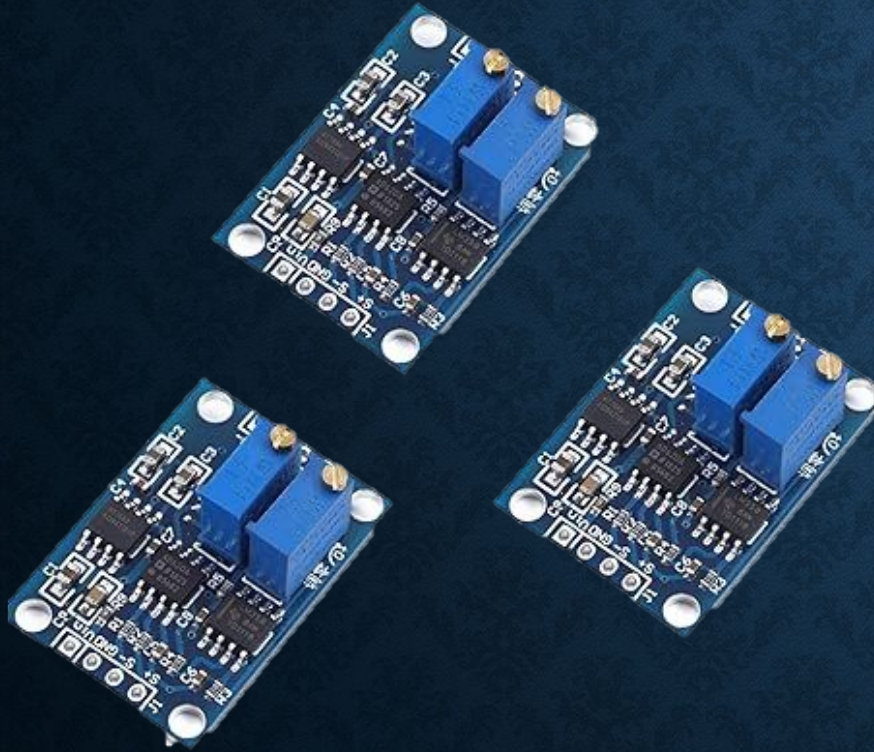


THE SOLUTION

After diagnosing the failure, we identified that the core issue was the common-mode noise – a problem that instrumentation amplifiers are specifically designed to combat. We decided to go with the AD620.

Why the AD620:

- Gain set by a single external resistor , making it simple, precise, and repeatable
- Excellent common-mode rejection ratio (CMRR), directly addressing our noise problem
- Far cleaner front-end performance for microvolt level signals compared to our op-amp chain
- Minimal external components reduced the chance of additional noise being introduced.



[AD620 Datasheet](#)

Result: Switching to the AD620 gave us a significantly cleaner signal and allowed the rest of the signal processing chain to function as intended.

| Frequency Detection Circuit

SIGNAL CHAIN



HOW IT WORKS

- ❌ Proton precession magnetometers measure Earth's magnetic field by detecting the **Larmor precession frequency** of hydrogen protons in a sensor fluid.
- ❌ The precession frequency is **directly proportional** to field strength via the proton gyromagnetic ratio ($\gamma_p \approx 42.577 \text{ MHz/T}$).
- ❌ For Earth's field ($\sim 25\text{--}65 \mu\text{T}$), the precession signal is typically **1,000–2,800 Hz** — a low-frequency, short-duration signal lasting only 1–3 seconds.
- ❌ Measuring this frequency with precision **better than 0.04 Hz** yields **sub-nanotesla** magnetic field resolution.

❌ Core Engineering Challenge

Accurately measuring a **low-frequency, short-duration, exponentially decaying** sinusoidal signal with the precision needed for scientific-grade magnetometry.

1–2.8

KHZ SIGNAL

<0.04

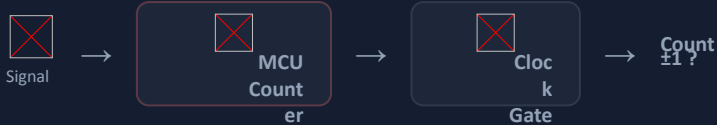
HZ PRECISION NEEDED

<1

NT RESOLUTION

Limitations of Microcontrollers & DFT

MICROCONTROLLER TIMER-BASED COUNTING



- ❌ Timer resolution limited by clock accuracy — in practice microcontrollers yielded **only 1 – 20 Hz accuracy**
- ❌ **±1 count ambiguity** at gate boundaries introduces significant error at low frequencies
- ❌ Timing jitter from **interrupt latency** and instruction execution degrades precision
- ❌ Cannot achieve better than **~0.4 Hz** precision for 1–3 kHz signals in practical gate times

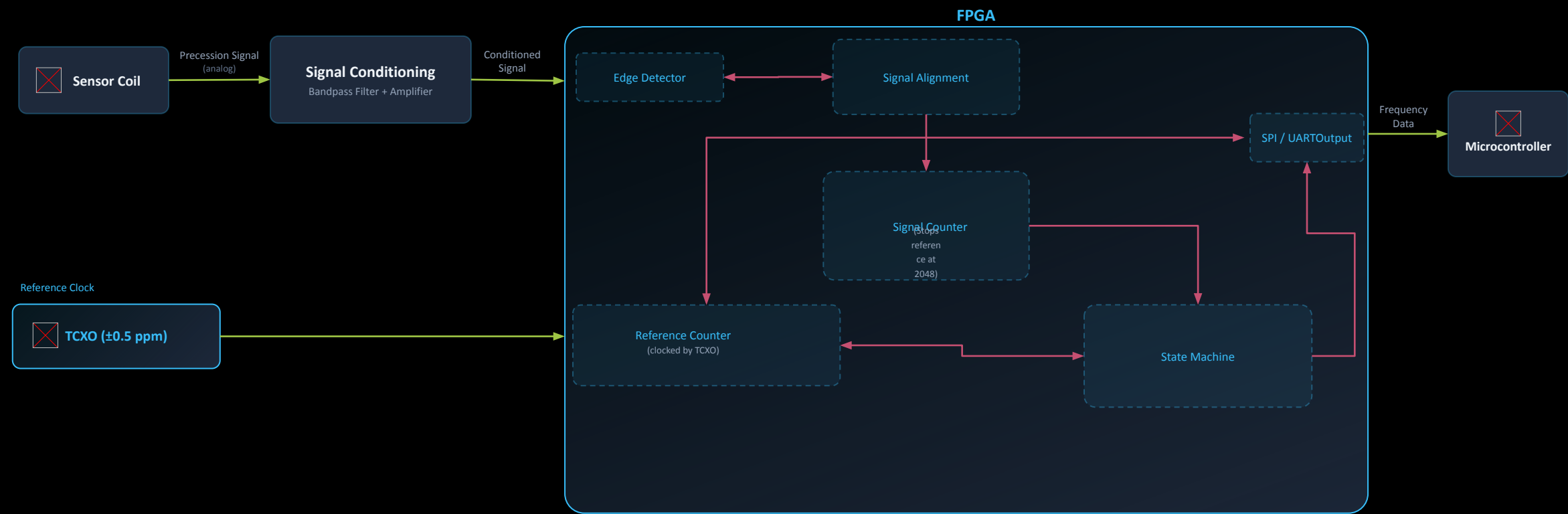
DFT / FFT SPECTRAL ANALYSIS



- ❌ Frequency resolution $\Delta f = 1/T$ — a 1-second signal yields only **1 Hz** resolution
- ❌ **Spectral leakage** from the decaying, windowed signal smears energy across bins
- ❌ Zero-padding improves interpolation but **not true resolution**
- ❌ **Computational cost** is high for real-time embedded processing
- ❌ 0.04 Hz resolution would require **25 seconds** of measurements

Neither approach reliably achieves the **<0.04 Hz** precision needed for sub-nanotesla magnetometry.

Design Choice — FPGA + TCXO Frequency Counter

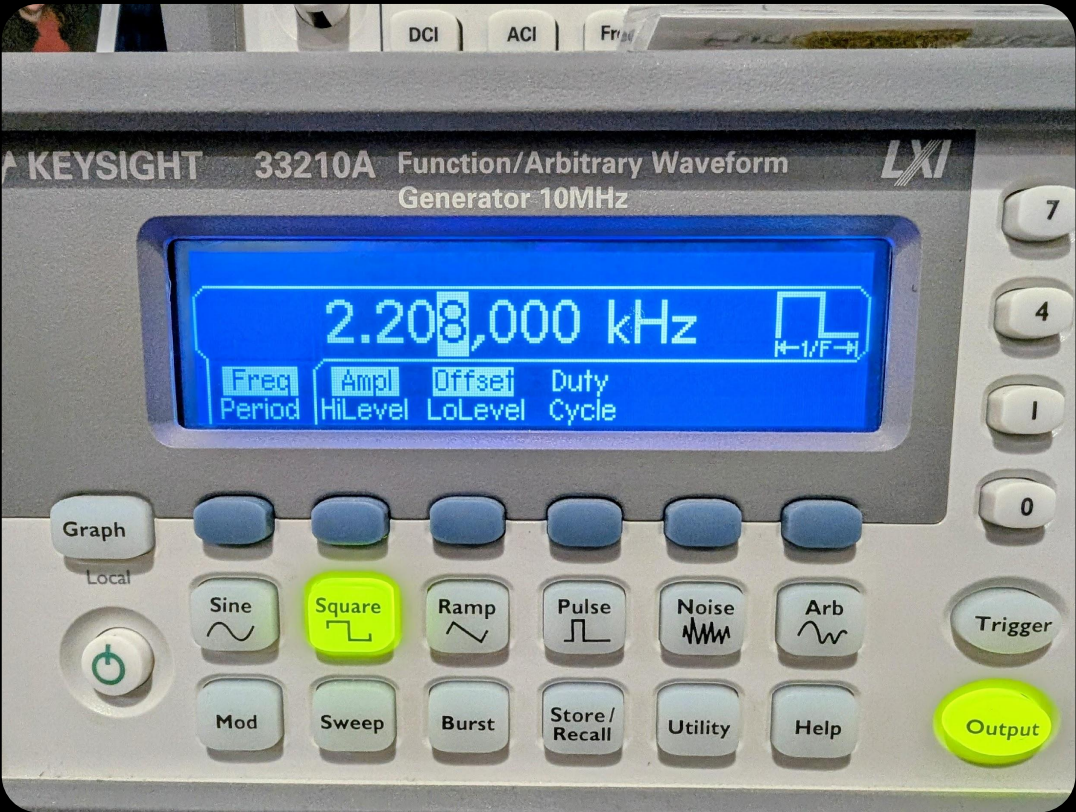


KEY ADVANTAGES

- TCXO provides a stable, low-drift reference clock (± 0.5 ppm or better)
- Reciprocal counting eliminates ± 1 count error by measuring period precisely
- FPGA enables true single-cycle counting with zero interrupt latency
- Theoretically achieves < 0.005 Hz frequency precision — enabling sub-nanotesla field resolution

Demo – Outputs from system

Frequency Input: 2208 Hz

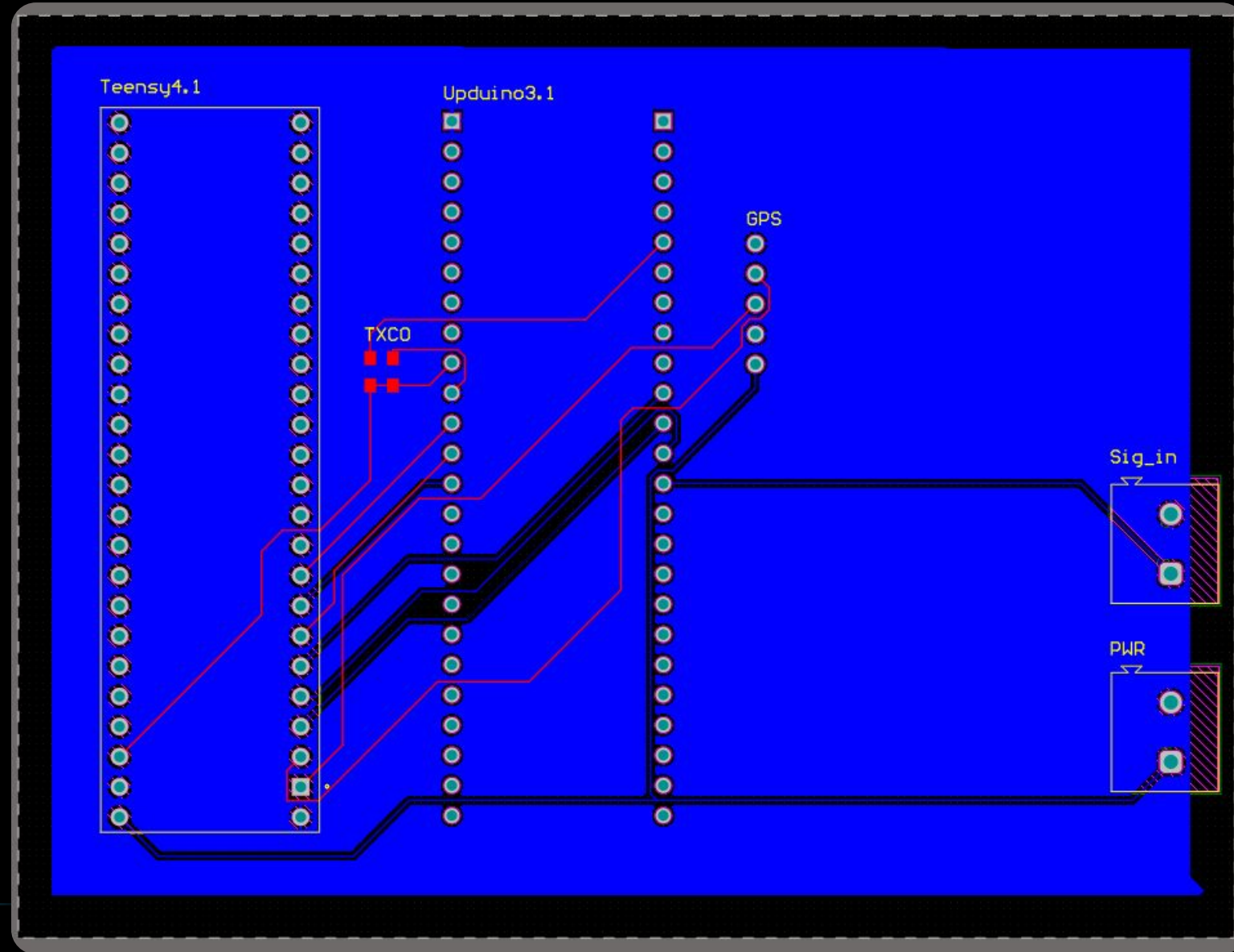


Frequency Output: 2207.99 Hz

104932	2207.99	NA	NA	NA	NA	NA	NA	NA
106865	2207.99	NA	NA	NA	NA	NA	NA	NA
108799	2207.99	NA	NA	NA	NA	NA	NA	NA
110733	2207.99	NA	NA	NA	NA	NA	NA	NA
112667	2207.99	NA	NA	NA	NA	NA	NA	NA
114601	2207.99	NA	NA	NA	NA	NA	NA	NA
116535	2207.99	NA	NA	NA	NA	NA	NA	NA
118469	2207.99	NA	NA	NA	NA	NA	NA	NA
120403	2207.99	NA	NA	NA	NA	NA	NA	NA
122337	2207.99	NA	NA	NA	NA	NA	NA	NA
124270	2207.99	NA	NA	NA	NA	NA	NA	NA
126204	2207.99	NA	NA	NA	NA	NA	NA	NA
128138	2207.99	NA	NA	NA	NA	NA	NA	NA
130072	2207.99	NA	NA	NA	NA	NA	NA	NA
132006	2207.99	NA	NA	NA	NA	NA	NA	NA
133940	2207.99	NA	NA	NA	NA	NA	NA	NA

Note: GPS is functional, but does not work indoors

Frequency detection and GPS PCB Design



HOW TO BUILD IT

(The frequency module)

IMPORTANT INFORMATION

- This module is designed to take in a 3.3V square wave.
- The input must not exceed 3.3V or the FPGA will be fried.
- It will save the GPS data and the frequency to a .CSV file on the SD card. One more simple calculation must be done to obtain the actual magnetic field value.
- The board can theoretically be powered by applying 3.3V to the board's power input, but this has not been tested. (We have been using the USB power on the Teensy 4.1) For a more reliable power system, apply 5V to the Teensy Vin pin.
- There has been an issue with the FPGA occasionally getting out of sync with the Teensy, which results in repeating "ERROR: measurement timed out" messages. It is only occasional, and resetting the FPGA in the code after this happens solves the issue, though it should be looked into further.

BILL OF MATERIALS

- Teensy 4.1 Microcontroller: [Digikey](#)
- UPduino v3.1 FPGA: [TinyVision](#)
- 12MHz TCXO reference clock (O 12,0-JT32CT-A-K-3,3-LF): [DigiKey](#)
- Goouuu Tech GT-U7 GPS module: [Amazon](#)
- PCB manufacturing files: [GitHub](#)
- 2x screw terminal: [DigiKey](#)
- Any microSD card: [Here's one](#)

SETTING UP THE TEENSY

- Download the Teensy Bootloader app: <https://www.pjrc.com/teensy/loader.html>
- I used the Arduino IDE with the Teensy board added (easy setting change) to load the .ino file: [GitHub](#)

SETTING UP THE UPDUINO FPGA

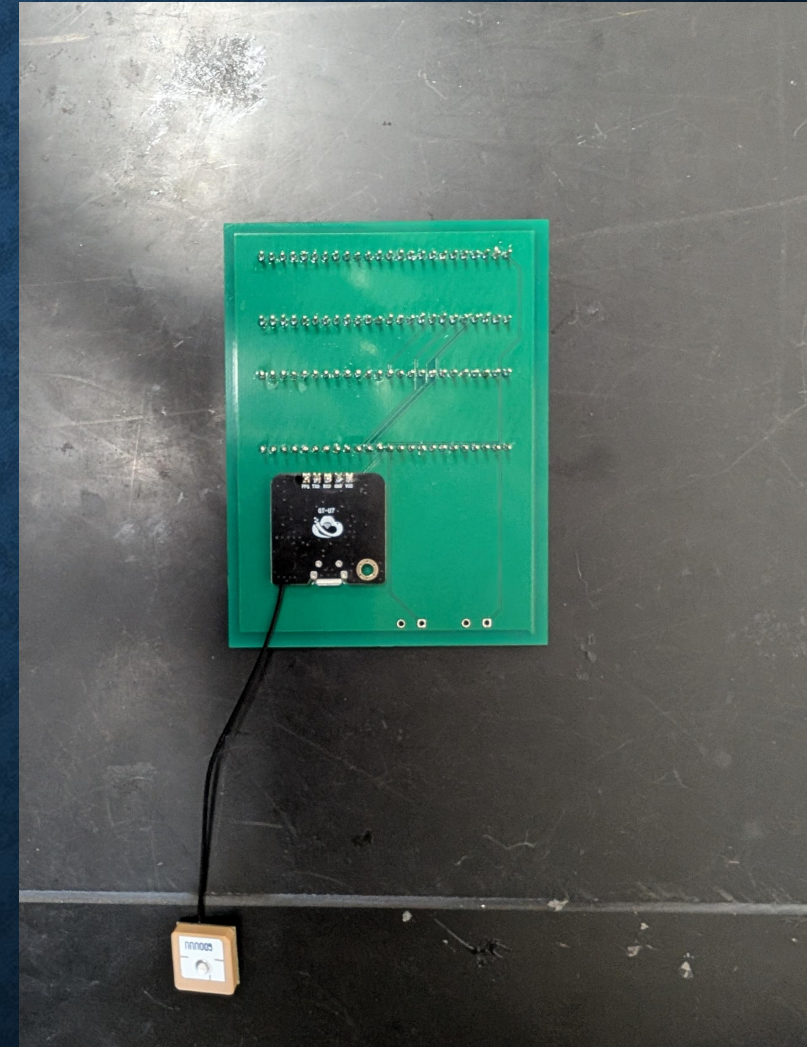
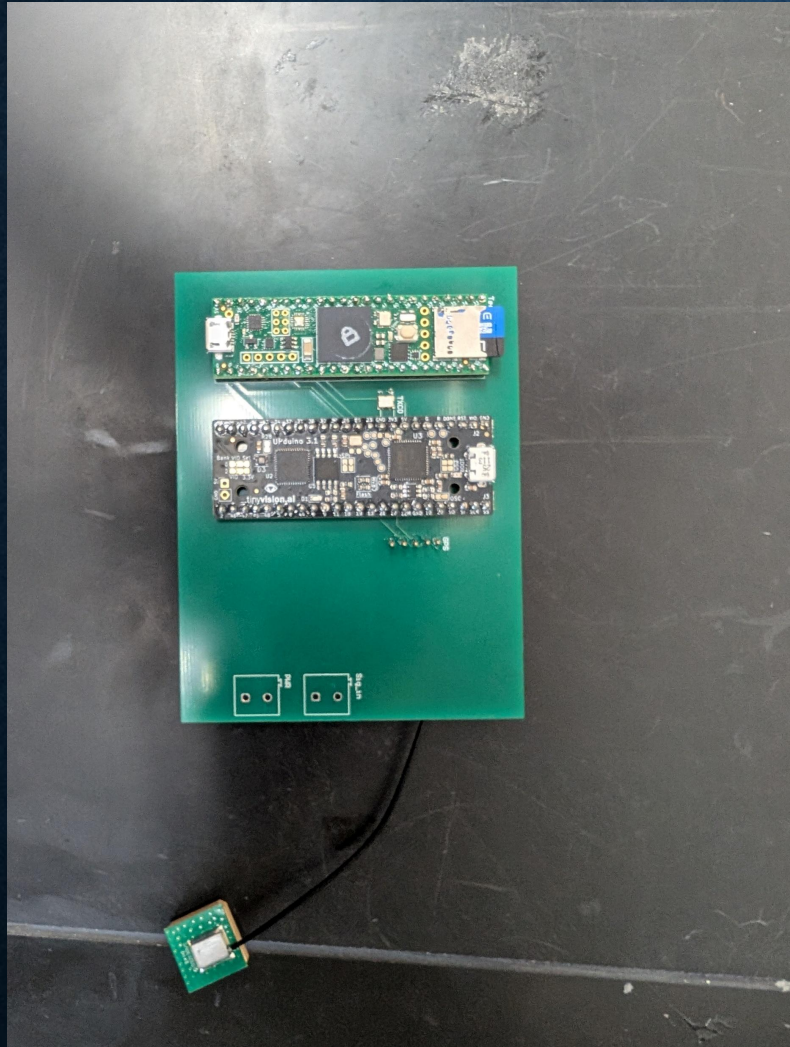
- Download the Icestorm toolchain and install it following the instructions listed in the GitHub: [IceStormToolchain](#)
- Either manually convert the .sv file and .pcf file and load the FPGA that way or download all the files in [this folder](#) and use the provided build.sh to load the FPGA. (The commands needed to do this manually are in the build.sh file)

BUILD THE BOARD

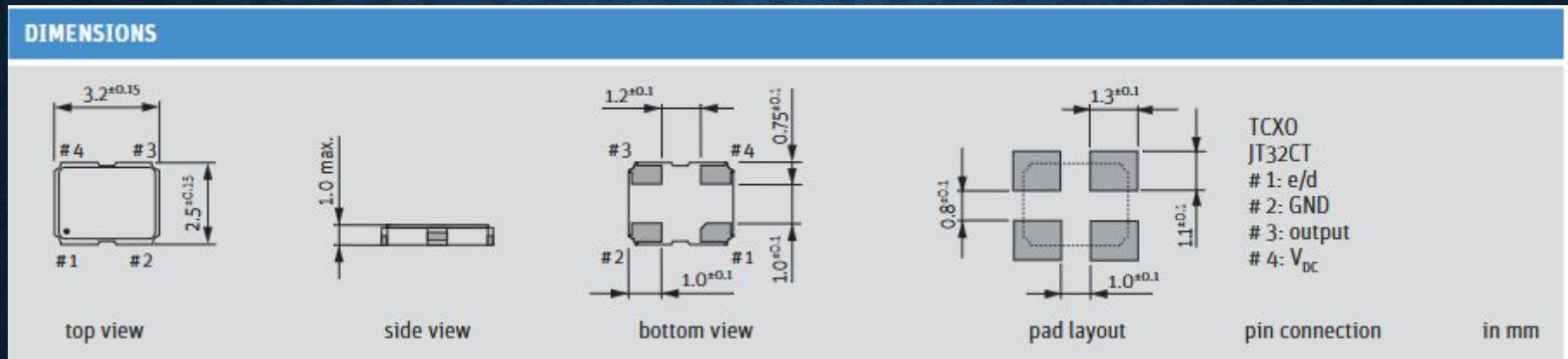
- If you put it together on the PCB, just solder everything together. (reference picture on the next slide)
- Otherwise, you must make these connections (as well as power connections):

Teensy 4.1	Upduino 3.1	TCXO	GPS module
0	-	-	RX
1	-	-	TX
2	13	-	-
3	19	-	-
4	21	-	-
5	25	-	-
6	26	-	-
7	23	-	-
-	20	3	-
-	18 to signal	-	-

ASSEMBLED PCB



TCXO DATASHEET CONNECTIONS



Pins #1 and #4 should be tied to 3.3V, #2 to GND and #3 to the FPGA as listed on the previous slide.

USE AND FUTURE CONSIDERATIONS FOR FREQUENCY CIRCUIT

- This unit takes in a conditioned signal and returns geo-tagged measurements on both the serial line and saved to the SD card.

Future considerations

- This unit should be integrated with the code from the polarization unit so that the Teensy can initiate the polarization and time it up with the GPS and frequency measurements.
- This unit should have voltage protection before the input signal reaches the FPGA (limit to 3.3V)
- Further investigations should be made into the syncing issue detailed on the IMPORTANT INFORMATION slide.
- 5V power should be supplied to the Teensy on the Vin pin, which can in turn power the FPGA and TCXO via it's 3.3V pin.

TOWFISH HOUSING UNIT

While we never ended up constructing the housing unit, we made a plan for how we were going to make it and the plan follows:

1. The Pipe:

- 8" Schedule 40 PVC Pipe
- We only need it to be a few feet long

2. Permanent End (glued shut):

- 8" PVC Schedule 40 End Cap (slip fit, solvent welded on permanently)

3. Removable End (the access side):

- 8" PVC Schedule 40 Female Cleanout Adapter (solvent welded onto pipe permanently)
- 8" PVC Schedule 40 Threaded Cleanout Plug (the removable piece)
- Rubber gasket or O-ring to go behind the plug face for the seal
- PVC solvent cement
- Teflon tape for the threads



CHALLENGES

Breadboard Noise: Breadboards have a noise floor that nearly masks the precession signal, making reliable measurement difficult. This significantly limits debugging and testing, making it challenging to assess overall system performance until components are implemented on a PCB.

Amplifier Spike Exposure: The amplifier currently lacks a timed analog switch to isolate it during coil excitation, exposing it to inductive voltage spikes that can saturate or damage the input stage. We have been working on a method to eliminate this issue.

Power Supply Ripple: Power supply ripple overlaps with the precession frequency (~ 2.1 kHz), introducing interference that contaminates the signal and reduces the SNR.

Magnetic Self-Interference: High-current switching in the polarization circuit generates unintended magnetic fields from nearby conductors, which couple into the sensor coil and distort the measured signal.

FUTURE PLANS

Resolving Current Challenges:

- Move all circuitry to PCB to eliminate breadboard noise floor
- Add timed analog switch to protect amplifier from inductive voltage spikes
- Separate power supply and signal ground planes to reduce ripple interference
- Physically isolate polarization circuit from sensor coils to reduce magnetic self-interference
- We will be working overtime to implement these changes by the final report deadline

Towfish Housing Unit:

- 8" PVC pipe enclosure with O-ring sealed custom end caps
- Waterproof cable glands for external connections
- Nose-mounted coil to minimize self-interference from onboard electronics
- Slide-out internal mount for easy SD card access and troubleshooting

THANK YOU!

All files from our project are available from our GitHub repository linked below.

<https://github.com/makeyes2/PPM-ECE455-SP26>